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Stressful life events and depressive symptom severity associated with suicide planning among adolescents with depression and non-suicidal self-injury: a cross-sectional study

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Abstract

Background Adolescent suicide is a major global public health concern. Suicide planning represents a critical transition in the suicidal process among adolescents with depression and non-suicidal self-injury (NSSI). Stressful life events and sleep disturbances are common in this population; however, their relative contributions to suicide planning remain unclear. This study aimed to examine the associations of stressful life events, insomnia symptoms, and emotional symptoms with suicide planning among adolescents with depression and NSSI.

Methods This cross-sectional study included 100 adolescents with depression and NSSI. Suicide planning within the past month was assessed using the suicide plan item (Q4) of the Mini-International Neuropsychiatric Interview. Stressful life events, insomnia symptoms, depressive symptoms, anxiety symptoms, and lifetime history of suicide attempt were assessed using clinical interview items and self-report measures. Multivariable logistic regression analyses were conducted to identify factors independently associated with suicide planning.

Results Among the participants, 57 (57.0%) reported suicide planning within the past month. Adolescents with suicide planning showed significantly greater exposure to stressful life events, higher insomnia symptom burden, and more severe depressive and anxiety symptoms compared with those without suicide planning. Stressful life events were significantly associated with insomnia symptom burden (standardized $\beta = 0.36$, $p < 0.001$). In the fully adjusted logistic regression model, stressful life events (OR = 1.07, 95% CI: 1.02–1.12, $p = 0.007$) and depressive symptoms (OR = 1.16, 95% CI: 1.02–1.31, $p = 0.019$) were independently associated with suicide planning, whereas insomnia symptoms were not independently associated with suicide planning.

Conclusions Among adolescents with depression and NSSI, stressful life events and depressive symptom severity are independently associated with suicide planning. These findings highlight the importance of addressing psychosocial stressors and depressive symptoms in suicide risk assessment and prevention strategies for high-risk adolescents.

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Keywords Suicide planning, Stressful life events, Depressive symptoms, Non-suicidal self-injury, Adolescents, Insomnia

Introduction

Suicide is a major global public health concern and was the third leading cause of death among individuals aged 15–29 years worldwide in 2021 [1]. Adolescents with depression who engage in non-suicidal self-injury (NSSI) represent a particularly vulnerable population, characterized by heightened emotional distress and substantially elevated risk for suicidal behaviors [2]. From a public health perspective, identifying modifiable factors associated with different stages of suicidality in this high-risk population is essential for improving suicide risk assessment and prevention strategies. Recent reviews have emphasized that adolescent suicide prevention requires a comprehensive approach involving early identification, mental health care access, targeted interventions, family and school engagement, and attention to modifiable clinical and psychosocial risk factors [3].

Within the suicidal process, suicide planning occupies a critical position, marking the transition from passive suicidal ideation to more organized and actionable intent. Compared with suicidal ideation alone, suicide planning has been shown to confer a substantially higher risk for subsequent suicide attempts and death by suicide [4–6]. However, many previous studies in adolescents have relied on composite or broad definitions of suicidality, which may obscure stage-specific correlates and mechanisms [7]. Focusing specifically on suicide planning may therefore provide greater clinical precision in identifying adolescents at imminent risk.

Stressful life events have been consistently implicated in the onset and exacerbation of depressive symptoms and suicidal behaviors among adolescents. Exposure to academic stress, interpersonal conflict, family adversity, and health-related stressors has been associated with increased suicidal ideation and attempts, particularly among youths with underlying emotional vulnerabilities or NSSI behaviors [8–11]. Depressive symptom severity is another well-established correlate of suicidality in adolescents [12, 13]. However, the relative and independent contributions of stress exposure and depressive symptoms to suicide planning remain incompletely understood, especially in high-risk clinical samples.

Sleep disturbances, particularly insomnia symptoms, are highly prevalent among adolescents with depression and NSSI and are closely linked to stress exposure and emotional dysregulation [14–16]. Prior studies have reported associations between insomnia and suicidal ideation or suicide attempts in adolescents and young adults [17–19]. However, findings have been mixed, and it remains unclear whether insomnia contributes independently to suicide planning after accounting for psychosocial stress and emotional symptoms, or whether sleep disturbance primarily reflects broader stress-related and affective burden [20, 21].

Accordingly, the present study aimed to examine the associations of stressful life events, insomnia symptoms, and emotional symptoms with suicide planning among adolescents with depression and NSSI. Specifically, we sought to (1) evaluate the relationship between stressful life events and insomnia symptom burden, and (2) identify factors independently associated with suicide planning after adjustment for depressive and anxiety symptoms and relevant demographic and clinical variables. We hypothesized that greater exposure to stressful life events and higher depressive symptom severity would be independently associated with suicide planning, whereas the association between insomnia symptoms and suicide planning would be attenuated after accounting for emotional factors.

Methods

Study design and participants

This was a cross-sectional observational study conducted among adolescents with depressive disorders and non-suicidal self-injury (NSSI). A total of 100 adolescents were consecutively recruited from the psychiatric outpatient clinic at Liuzhou Workers' Hospital. Diagnoses of depressive disorders were made by attending psychiatrists according to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) criteria [22] as part of routine clinical practice. Non-suicidal self-injury was classified according to the DSM-5 proposed diagnostic criteria.

Written informed consent was obtained from participants' parent(s) or legal guardian(s), and written assent was obtained from all adolescent participants prior to participation. The study protocol was approved by the local institutional ethics committee and conducted in accordance with the Declaration of Helsinki.

Assessment of suicide planning

Diagnoses of depressive disorders and eligibility related to NSSI were screened and confirmed by psychiatrists as part of the clinical assessment. Suicide planning within the past month was assessed using the suicide plan item (Q4) of the Mini-International Neuropsychiatric Interview (MINI). The MINI suicide-related items were administered as a structured clinical interview by trained

psychiatrists or therapists with experience in adolescent mental health assessment, rather than completed independently by participants. Participants who endorsed the presence of a suicide plan during the past month were classified as having suicide planning, whereas those who did not endorse this item were classified as having no suicide planning.

Stressful life events

Exposure to stressful life events was assessed using the Adolescent Life Events Scale, a self-report questionnaire designed to evaluate recent stressors commonly experienced by adolescents, including academic stress, interpersonal difficulties, family-related stress, and health-related events. A total life events score was calculated by summing all item scores, with higher scores indicating greater cumulative stress exposure [23]. In the present sample, internal consistency was good (Cronbach's $\alpha = 0.891$).

Insomnia symptoms

Insomnia symptom burden was assessed using an investigator-constructed sleep questionnaire. The insomnia symptom burden score was calculated by summing six symptom-related items during the past month: difficulty initiating sleep, difficulty maintaining sleep or frequent nocturnal awakening, early morning awakening with difficulty returning to sleep, use of sleep medication due to insomnia, distress related to insomnia, and functional impairment in daily life, work, or study due to insomnia. Higher scores indicated greater insomnia symptom burden. The individual items are provided in Supplementary Table S1. In the present sample, the internal consistency of this composite was acceptable (Cronbach's $\alpha = 0.761$).

Depressive and anxiety symptoms

Depressive symptoms were assessed using the 13-item short form of the Beck Depression Inventory (BDI-SF) [24, 25]. Each item was scored from 0 to 3, and total scores were calculated by summing all 13 items. Anxiety symptoms were assessed using the Beck Anxiety Inventory (BAI) [26]. Both instruments are widely used self-report measures, with higher total scores indicating greater symptom severity. In the present sample, internal consistency was good for the BDI-SF (Cronbach's $\alpha = 0.883$) and excellent for the BAI (Cronbach's $\alpha = 0.918$).

Lifetime history of suicide attempt

Lifetime history of suicide attempt was assessed using the corresponding item of the MINI and coded as a dichotomous variable (yes/no). One participant had missing data for this variable.

Demographic variables

Demographic information including age (years) and sex (female/male) was collected for all participants.

Statistical analysis

Descriptive statistics were calculated for all study variables. Continuous variables are presented as means and standard deviations (SD), and categorical variables are presented as counts and percentages. Group differences between adolescents with and without suicide planning were examined using independent-samples t tests for continuous variables and χ^2 tests or Fisher's exact tests for categorical variables, as appropriate.

To examine the association between stressful life events and insomnia symptoms, linear regression analysis was conducted with insomnia symptom burden as the dependent variable and life events total score as the independent variable.

Factors associated with suicide planning were examined using multivariable logistic regression analysis, with suicide planning (yes/no) as the dependent variable. Independent variables included stressful life events total score, insomnia symptom burden, depressive symptoms (BDI-SF), anxiety symptoms (BAI), age, sex, and lifetime history of suicide attempt. All variables were entered simultaneously into the model. Results are reported as odds ratios (ORs) with 95% confidence intervals (CIs). Model performance was further evaluated using the omnibus model test, Nagelkerke R^2, the Hosmer–Lemeshow goodness-of-fit test, overall classification accuracy, and the area under the receiver operating characteristic curve (AUC). Internal consistency of multi-item self-report instruments in the present sample was assessed using Cronbach's alpha.

All analyses were performed using IBM SPSS Statistics (version 30.0; IBM Corp., Armonk, NY, USA). A two-tailed p value < 0.05 was considered statistically significant.

Results

Participant characteristics

A total of 100 adolescents with depression and non-suicidal self-injury were included in the analysis. Of these, 57 (57.0%) reported suicide planning within the past month.

As shown in Table 1, adolescents with suicide planning reported significantly greater exposure to stressful life events, higher insomnia symptom burden, and more severe depressive and anxiety symptoms compared with those without suicide planning (all $p < 0.05$). In addition, the proportion of participants with a lifetime history of suicide attempt was significantly higher among those with suicide planning. There were no significant between-group differences in age or sex.

Table 1 Participant characteristics by suicide planning status

Characteristic	No suicide plan ($n=43$)	Suicide plan ($n=57$)	p value
Age, years	14.95 ± 1.21	14.72 ± 1.39	0.38
Sex, female, n (%)	41 (95.3%)	49 (86.0%)	0.18
Life events total score	63.27 ± 15.30	82.06 ± 19.33	<0.001
Insomnia symptom burden	6.95 ± 1.60	7.54 ± 1.34	0.048
BDI-SF total score	20.23 ± 7.13	25.70 ± 6.44	<0.001
BAI total score	24.02 ± 9.91	32.11 ± 12.48	<0.001
Lifetime suicide attempt, n (%)	8(19.05%)	24(42.11%)	0.015

Values are presented as mean ± standard deviation (SD) for continuous variables and number (percentage) for categorical variables. Suicide planning was defined based on endorsement of a suicide plan within the past month (Q4). Group comparisons were conducted using independent-samples t tests for continuous variables and χ^2 tests or Fisher's exact tests for categorical variables, as appropriate. A two-tailed p value < 0.05 was considered statistically significant

Table 2 Association between life events and insomnia symptom burden

Predictor	B (SE)	Standardizedβ	p value	R^2
Life events total score	0.026 (0.008)	0.358	< 0.001	0.128

Results are derived from a linear regression model with insomnia symptom burden as the dependent variable. B indicates the unstandardized regression coefficient; SE, standard error. R^2 represents the proportion of variance explained by the model. A two-tailed p value < 0.05 was considered statistically significant

Association between life events and insomnia symptoms

Linear regression analysis demonstrated a significant positive association between stressful life events and insomnia symptom burden (Table 2). Greater exposure to stressful life events was associated with more severe insomnia symptoms (B = 0.026, SE = 0.008, standardized β = 0.358, $p < 0.001$). Stressful life events explained 12.8% of the variance in insomnia symptom burden ($R^2 = 0.128$).

Factors associated with suicide planning

Results of the multivariable logistic regression analysis examining factors associated with suicide planning are presented in Table 3.

After simultaneous adjustment for insomnia symptoms, stressful life events, depressive symptoms, anxiety symptoms, age, sex, and lifetime history of suicide attempt, stressful life events (OR = 1.07, 95% CI: 1.02–1.12, $p = 0.007$) and depressive symptoms (OR = 1.16, 95% CI: 1.02–1.31, $p = 0.019$) remained independently associated with suicide planning.

In contrast, insomnia symptom burden was not independently associated with suicide planning in the fully adjusted model (OR = 1.16, 95% CI: 0.72–1.86, $p = 0.543$). Anxiety symptoms, age, sex, and lifetime history of suicide attempt were also not significantly associated with suicide planning after adjustment. The overall logistic regression model was statistically significant, $\chi^2(7) = 33.374$, $p < 0.001$. The model explained 44.0% of the variance in suicide planning based on Nagelkerke R^2 and showed acceptable fit according to the Hosmer–Lemeshow goodness-of-fit test, $\chi^2(8) = 9.123$, $p = 0.332$. The overall classification accuracy was 78.6%, and the area under the ROC curve was 0.841 (95% CI: 0.757–0.925, $p < 0.001$).

Sensitivity analyses using recent suicide attempt as an alternative outcome yielded broadly consistent results (Supplementary Table S2). Sensitivity analysis using the BDI-SF total score with the suicide/self-harm item excluded yielded broadly consistent results (Supplementary Table S3). Stressful life events remained independently associated with suicide planning (OR = 1.07, 95% CI: 1.02–1.12, $p = 0.008$), and depressive symptoms excluding the suicide/self-harm item also remained significant (OR = 1.14, 95% CI: 1.01–1.30, $p = 0.037$). Insomnia symptoms remained non-significant (OR = 1.16, 95% CI: 0.73–1.86, $p = 0.527$).

Discussion

In this cross-sectional study of adolescents with depression and NSSI, we examined factors associated with suicide planning, a critical stage in the suicidal process. The principal findings were that exposure to stressful life events and depressive symptom severity were

Table 3 Multivariable logistic regression analysis of factors associated with suicide planning among adolescents with depression and NSSI

Predictor	B	SE	OR (Exp[B])	95% CI for OR	p value
Insomnia symptom score	0.147	0.241	1.16	0.72–1.86	0.543
Life events total score	0.066	0.025	1.07	1.02–1.12	0.007
Depressive symptoms (BDI-SF)	0.148	0.063	1.16	1.02–1.31	0.019
Anxiety symptoms (BAI)	−0.063	0.046	0.94	0.86–1.03	0.176
Age (years)	−0.074	0.226	0.93	0.60–1.45	0.743
Female sex	−1.872	1.052	0.15	0.02–1.21	0.075
Lifetime suicide attempt	0.678	0.678	1.97	0.52–7.44	0.318
Constant	-4.463	3.921	—	—	0.255

Abbreviations: *OR*, odds ratio; *CI*, confidence interval; *BDI-SF*, 13-item short form of the Beck Depression Inventory; *BAI*, Beck Anxiety Inventory. Suicide planning was defined as endorsement of a suicide plan during the past month (MINI item Q4). All variables were entered simultaneously into the logistic regression model

independently associated with suicide planning, whereas insomnia symptoms were not independently associated after adjustment for emotional symptoms and other relevant clinical factors. In addition, stressful life events were significantly associated with insomnia symptom burden, suggesting a close link between stress exposure and sleep disturbance in this high-risk population. Within this clinically high-risk sample, these findings highlight the importance of psychosocial stress and depressive symptoms in identifying adolescents at elevated risk for suicide planning [3].

Stressful life events and suicide planning

Stressful life events emerged as the most robust and consistent factor associated with suicide planning. This finding is consistent with stress-diathesis models of suicidality, which posit that acute or cumulative psychosocial stressors can precipitate suicidal processes in vulnerable individuals [27, 28]. Adolescents with depression and NSSI may be particularly sensitive to environmental stressors, such as academic pressure, interpersonal conflict, and family-related adversity, which can overwhelm coping capacities and contribute to the escalation from suicidal ideation to more organized planning [29, 30]. By focusing specifically on suicide planning rather than broader or composite suicidal outcomes, the present findings underscore the importance of stress exposure in this critical transition stage.

Depressive symptoms as an independent correlate

Depressive symptom severity was independently associated with suicide planning even after accounting for stress exposure, insomnia symptoms, and other covariates. This result is consistent with extensive prior evidence identifying depression as a central clinical correlate of suicidality in adolescents [12, 31]. Beyond general emotional distress, depressive symptoms may amplify the impact of stressors on suicide planning through mechanisms such as hopelessness, cognitive constriction, and impaired problem-solving [27, 28]. The independent contribution of depressive symptoms observed in the present study highlights the need for careful assessment and targeted treatment of depression in adolescents with NSSI when evaluating suicide planning risk.

Insomnia symptoms and their attenuated association

Although insomnia symptoms were more severe among adolescents with suicide planning and were significantly associated with stressful life events, they were not independently associated with suicide planning in the fully adjusted model. Given the cross-sectional design, this finding should not be interpreted as evidence that insomnia symptoms are causally downstream of stress or depressive symptoms. Rather, the attenuation of the association after adjustment suggests that the relationship between insomnia symptoms and suicide planning may partly reflect shared variance with psychosocial stress exposure and depressive symptom severity.

Previous studies have reported mixed findings regarding the role of sleep disturbance in suicidality, with some demonstrating independent associations and others reporting attenuation after controlling for affective symptoms [19, 20, 31]. In adolescents with depression and NSSI, insomnia symptoms may represent a clinical marker of broader stress-related and affective burden rather than an independent correlate of suicide planning. This hypothesis requires confirmation in longitudinal studies capable of clarifying temporal and potentially mediating relationships.

Clinical implications

The present findings have several important clinical implications. First, they underscore the importance of systematically assessing stressful life events when evaluating suicide planning risk in adolescents with depression and NSSI, as stress exposure appears to play a central role in this critical stage of suicidality [8, 10]. Second, the independent association between depressive symptoms and suicide planning highlights depression severity as a key therapeutic target in suicide prevention efforts [12, 31]. Third, although insomnia symptoms were not independently associated with suicide planning after adjustment, their association with stress exposure suggests that sleep disturbances may serve as a clinically accessible marker of broader psychosocial and affective burden [14, 16]. Clinically, these findings support a comprehensive risk assessment framework that incorporates recent stressful life events, depressive symptom severity, sleep disturbance, suicidal ideation, and prior suicidal behavior among adolescents with depression and NSSI. These findings are also consistent with recent calls for comprehensive adolescent suicide prevention strategies that integrate early identification, targeted mental health interventions, and attention to modifiable psychosocial risk factors [2, 3].

Limitations

Several limitations should be considered when interpreting these findings. First, the cross-sectional design precludes causal or temporal inferences regarding the relationships among stressful life events, insomnia symptoms, depressive symptom severity, and suicide planning. Therefore, the observed associations should be interpreted as correlational. In particular, the possibility that insomnia symptoms may reflect broader stress-related or affective burden should be regarded as a hypothesis rather than evidence of a temporal or mediating pathway.

Second, suicide planning was assessed using a single dichotomous item from the MINI. Although this approach is commonly used in structured clinical interviews and provides a clinically interpretable indicator of recent suicide planning, it may oversimplify a complex and heterogeneous phenomenon. Suicide planning may vary in specificity, lethality, feasibility, access to means, and temporal proximity. Therefore, classifying suicide planning as a binary outcome may have obscured clinically meaningful heterogeneity among adolescents who endorsed a suicide plan. Future studies should consider using more detailed assessments of suicide planning to capture these dimensions.

Third, although the regression model adjusted for several demographic and clinical variables, residual confounding cannot be excluded. Important factors such as psychiatric comorbidities beyond depression and anxiety, medication use, family psychiatric history, treatment status, and recent acute stressors were not assessed or not included in the model. These factors may influence both stress exposure or sleep disturbance and suicide planning, thereby limiting the interpretability of the observed associations. Future studies with larger samples and more comprehensive clinical assessments are needed to better account for these potential confounders.

Fourth, the sample was relatively small and clinically specific, consisting of adolescents with both depression and NSSI recruited from an outpatient psychiatric clinic. The high prevalence of suicide planning in this sample (57.0%) indicates that the participants represented a particularly high-risk subgroup. Therefore, the findings should be interpreted primarily in relation to adolescents with depression and NSSI in clinical outpatient settings and may not be generalizable to community samples, adolescents with depression but without NSSI, or broader adolescent populations. In addition, the relatively small sample size in relation to the number of predictors included in the logistic regression model resulted in approximately eight outcome events per variable, which falls slightly below the commonly recommended threshold of ten events per variable. The highly imbalanced sex distribution and the small number of male participants may also have contributed to potential quasi-complete separation for the sex variable, thereby limiting the precision and stability of individual coefficient estimates.

Fifth, insomnia symptoms were assessed using an investigator-constructed composite rather than a formally validated insomnia scale. Although the composite showed acceptable internal consistency in the present sample, future studies should use standardized insomnia measures and, where feasible, objective sleep assessments. In addition, although the sensitivity analysis excluding the BDI-SF suicide/self-harm item yielded broadly consistent results, the conceptual overlap between depressive symptom severity and suicidality should still be considered when interpreting the association between BDI-SF scores and suicide planning. Finally, all symptom measures were based on self-report, which may be subject to reporting bias.

Future studies using longitudinal designs and multimethod assessments, including objective sleep measures, are needed to clarify the temporal relationships among stress exposure, sleep disturbance, and suicidality.

Conclusions

Among adolescents with depression and NSSI, stressful life events and depressive symptom severity are independently associated with suicide planning. These findings highlight the importance of addressing psychosocial stressors and depressive symptoms when assessing suicide risk in high-risk adolescents and may inform prevention and early intervention strategies targeting suicide planning.

Abbreviations

BDI-SF	13-item short form of the Beck Depression Inventory
BAI	Beck Anxiety Inventory
DSM-5	Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition
MINI	Mini-International Neuropsychiatric Interview
NSSI	Non-suicidal self-injury

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-026-28151-0.

Supplementary Material 1.

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Authors' contributions

Xiaoyan He conceived and designed the study. Rui Wang, Yan Chen, Caihong Lin, Jiaqi Mo and Yanqiu Wei collected the data. Tingting Hong and Ronghong Gao performed the statistical analyses. LiLi Guo and Jie Li interpreted the data . LiLi Guo and Xiaoyan He drafted the manuscript. All authors critically revised the manuscript for important intellectual content and approved the final version.

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Data availability

The datasets generated and/or analysed during the current study are not publicly available due to the sensitive nature of clinical data and the potential risk to participant privacy, but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Committee of Liuzhou Workers' Hospital (approval number: KY2020040). The study was conducted in accordance with the Declaration of Helsinki and relevant local regulations. Written informed consent was obtained from participants' parent(s) or legal guardian(s), and written assent was obtained from all adolescent participants prior to participation.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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